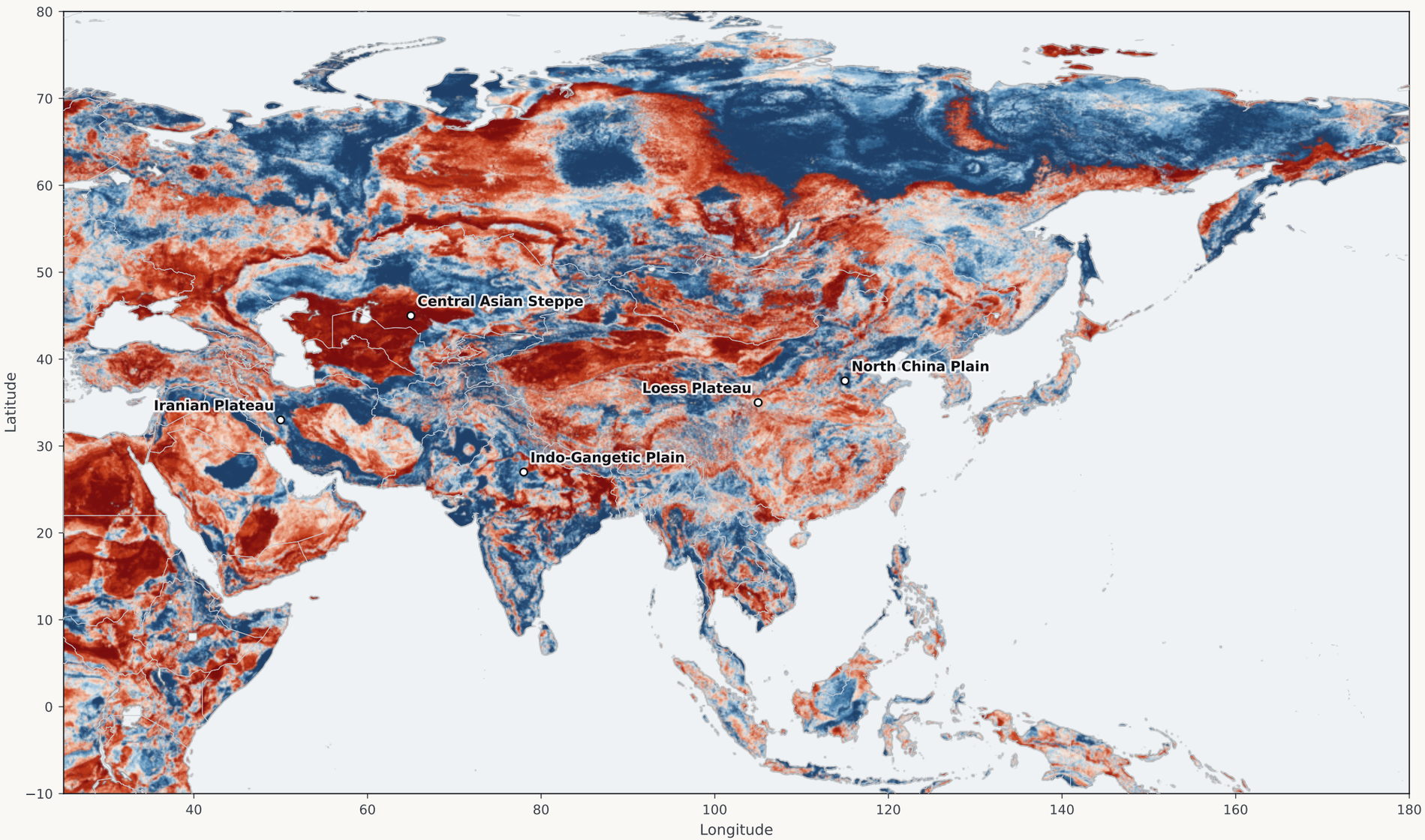
# Climate-Corrected Soil Microbial Respiration Anomaly

*Predicted soil respiration ÷ climate-expected soil respiration. Values below 1.0 indicate microbial activity is suppressed relative to what climate alone would support — a signal of soil-driven degradation.*




ANOMALY RATIO

0.50 0.75 1.00 1.25 1.50

INTERPRETATION

| | |
|---|---|
| **≥ 1.25** | Microbial activity exceeds climate expectation |
| **1.00 – 1.25** | Healthy / functioning soils |
| **≈ 1.00** | At climate expectation |
| **0.75 – 1.00** | Mild suppression |
| **≤ 0.75** | Significant degradation — addressable by management |

MODEL XGBoost · spatial-block CV · resolution ~5 km

DATA COSORE + SRDB soil respiration · SoilGrids 2.0 · WorldClim 2.1 · MODIS

CV $R^2$ — Asia→US $R^2$ —

N(train)=615 N(US)=— 2026-04-26